(ii) 49^{17}

Unit digit of base 49 is 9 and power is 17 (odd power).

Therefore, unit digit of 49^{17} is 9.

The following activity will help you to find the unit digits of exponent numbers whose base is ending with 2,3,7 and 8.



Activity

Observe the table given below. The numbers in first column, that is 2,3,7 and 8 denotes the unit digit of base of the given exponent number and the numbers in the first row, that is 1,2,3 and 0 stands for the remainder when power is divided by 4.

Unit digit of base	The remainder when power is divided by 4			
	1	2	3	0
2	2	4	8	6
3	3	9	7	1
7	7	9	3	1
8	8	4	2	6

For example, consider 2^6

Unit digit of base 2 is 2 and power is 6. When the power 6 is divided by 4, we get the remainder as 2.

From the table, we see that 2 and 2 corresponds to 4. Therefore, the unit digit of 2^6 is 4. To verify, $2^6 = 2\times2\times2\times2\times2\times2 = 64$.

Similarly, consider 117^{20}

Unit digit of base 117 is 7 and power is 20. When the power 20 is divided by 4, we get the remainder as 0.

From the table, we see that 7 and 0 corresponds to 1. Therefore, the unit digit of 117^{20} is 1.

Now, you can extend this activity for finding the unit digits of any exponent number whose base is ending with any of 2,3,7 or 8.

Exercise 3.2

1. Fill in the blanks.

   (i) Unit digit of $124\times36\times980$ is _____

   (ii) When the unit digit of the base and its expanded form of that number is 9, then the exponent must be _____ power.

7th_Maths_T2_EM_Chp3.indd 56 09-07-2019 12:20:55